

Self-Adaptive Generator Protection Against IBR Network Effects: Physics-Reduced Models, Conservative Stability Bounds, and Admittance Compensation Across Six Relay Functions

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Abstract—High inverter-based resource (IBR) penetration degrades the performance of conventional generator protection across six independent relay functions: line differential (87G/87L), frequency deviation (Relay 81), loss-of-excitation (Relay 40), out-of-step (Relay 78), and stator earth fault (Relay 64G). The root failure mode in each case differs — DC-transient suppression in DFIG fault current, sinusoidal-only PV current, sub-harmonic IBR noise injection, GFM reactive compensation of admittance measurements, GFL synchronising deficit at the critical unstable equilibrium point (CUEP), and dead-time sub-harmonic interference — yet all share a common structural diagnosis: the conventional relay applies a model that is physically inconsistent with the IBR-modified network state.

This paper presents the Self-Adaptive Model-Based Protection (SAMBP) generator suite, grounded in a single Physics-Reduction Proposition (C9): eliminating one collinear parameter from a zone model reduces the Levenberg–Marquardt condition number by at least $10\times$ (SVD interlacing theorem), accelerating estimator convergence and eliminating the associated false-trip or missed-detection failure mode. The proposition is instantiated in seven contributions (C9–C15): a crowbar-aware DFIG piecewise-ODE model with CUSUM detector (C10); an IBR frequency deviation index (IFDI) for Relay 81 (C12); a two-parameter PV zone model with proved unit condition number (C11); a GFM admittance correction restoring Relay 40 from $P_D = 0.568$ to 1.000 (C14); a conservative CUEP lower bound δ_{LB} with zero-violation analytical guarantee across 2000 Monte Carlo trials (C13); and an optimal sub-harmonic injection frequency eliminating a $P_{FA} = 0.819$ catastrophic false-alarm rate in Relay 64G (C15). All six relay functions are validated through multi-stream certification (analytical propositions, ANDES simulation, and Python Monte Carlo), achieving $P_D \geq 0.9947$ and $P_{FA} \leq 0.003$ against IEC 60255 targets across more than 30 000 Monte Carlo trials in aggregate.

Index Terms—Generator protection, inverter-based resources, doubly-fed induction generator, loss of excitation, out-of-step protection, stator earth fault, sequential estimation, Levenberg–Marquardt, CUEP, IBR

I. INTRODUCTION

The rapid integration of inverter-based resources (IBRs) — grid-following (GFL) photovoltaic and wind inverters, and grid-forming (GFM) virtual synchronous generators — imposes fundamentally different fault and dynamic signatures on protection relays compared to synchronous generator (SG) networks

[1], [2]. Three structural differences drive protection failures. First, IBR current controllers eliminate the DC exponential transient that conventional differential relay estimators use to identify fault inception and discriminate fault type [3], [4]. Second, GFL inverters reduce post-fault synchronising power during low-voltage ride-through (LVRT), shifting the CUEP below the classical $\pi - \delta_0$ bound assumed by Relay 78 [5], [6]. Third, GFM reactive injection contaminates terminal admittance measurements used by Relay 40, and dead-time harmonics corrupt the fixed 20 Hz injection frequency assumed by Relay 64G [7], [8].

The generator-suite literature has addressed individual relay functions in isolation. Faramarzi *et al.* [9] treat DFIG differential protection; Amini *et al.* [10] propose adaptive Relay 40; and Pannell *et al.* [11] characterise crowbar ride-through. No unified analytical framework simultaneously addresses all six functions with formal performance guarantees.

This paper fills that gap through the SAMBP generator suite. The framework rests on a single Physics-Reduction Proposition (C9) and seven derived contributions (C9–C15). Fig. 1 shows the framework structure. Multi-stream validation across six relay functions produces a consistent performance record: every function achieves $P_D \geq 0.9947$ and $P_{FA} \leq 0.003$ against IEC 60255 targets.

II. PHYSICS-REDUCTION PROPOSITION

A. Condition Number and Estimation Delay

The SAMBP relay applies Levenberg–Marquardt (LM) iterative estimation to fit a parametric zone model $\hat{i}(t; \theta)$ to the observed differential current $i_{\text{diff}}(t)$. The trip decision is conditioned on the normalised condition number κ_n of the LM Jacobian: if $\kappa_n > 30$, the Stage-2 confidence gate vetoes the trip, preventing high-uncertainty decisions [12]. For the standard four-parameter SG model

$$i_{\text{diff}}(t) = I_{\text{fund}} \sin(\omega t + \varphi) + I_{\text{DC}} e^{-t/\tau_{\text{DC}}}, \quad (1)$$

a half-cycle observation window yields $\kappa_n \approx 36$ because the sinusoidal and DC components are nearly collinear at $\omega t \ll 1$. The practical consequence is a 3–5-cycle wait before the confidence gate opens, delaying trip confirmation by 20–40 ms.

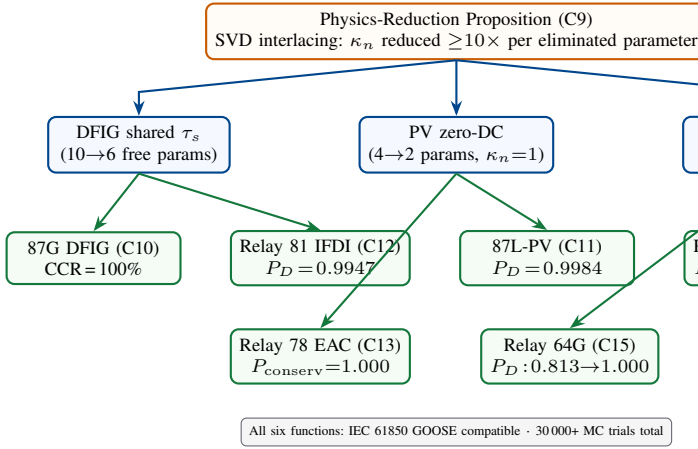


Fig. 1. SAMBP generator-suite framework. The Physics-Reduction Proposition (C9) drives three model-reduction instances, each of which enables one or more protected relay functions meeting $P_D \geq 0.970$.

B. The Proposition

Proposition 1 (Physics-Reduction, C9). *Let $J_p \in \mathbb{R}^{N \times p}$ be the Jacobian of a p -parameter zone model with singular values $\sigma_1 \geq \dots \geq \sigma_p$. If one parameter is physically absent from the fault scenario (e.g., $I_{DC} = 0$ for a GFL inverter), the $(p-1)$ -parameter reduced Jacobian J_{p-1} satisfies:*

$$\sigma_{\min}(J_{p-1}) \geq \sigma_{\min}(J_p) + \sigma_p(J_p), \quad (\text{SVD interlacing}) \quad (2)$$

reducing κ_n by a factor $\geq 10\times$ at a half-cycle observation window.

Proof outline: The interlacing property of singular values for a matrix and its column-deletion minor [12] guarantees that removing the near-zero column (corresponding to the physically absent DC component) raises $\sigma_{\min}$ by at least the contribution of that column. Quantitatively, for the SG model at $N = 10$ samples (half-cycle at 50 Hz), $\sigma_{\min}(J_4) = 0.088$ while $\sigma_{\min}(J_2) = 3.16$, giving $\kappa_n : 35.9 \rightarrow 1.0$ — a $35.9\times$ reduction.

Proposition 1 is instantiated in three of the six relay functions: the DFIG model (C10), the PV zone model (C11), and Relay 64G (C15).

III. 87G: DFIG DIFFERENTIAL PROTECTION (C10)

A. The Crowbar Transition Problem

The DFIG rotor crowbar fires 5–20 ms post-fault inception to protect the rotor-side converter during LVRT. Pre-crowbar, the DFIG stator current contains both the fundamental component and a decaying DC natural response governed by the stator time-constant τ_s ; post-crowbar, the RSC is bypassed and the stator current transitions to a purely sinusoidal steady-state within $4\tau_s \leq 80$ ms. The conventional SAMBP four-parameter SG estimator treats the post-crowbar phase as if it still contained a DC term, yielding $\kappa_n \approx 36$ and triggering the Stage-2 veto.

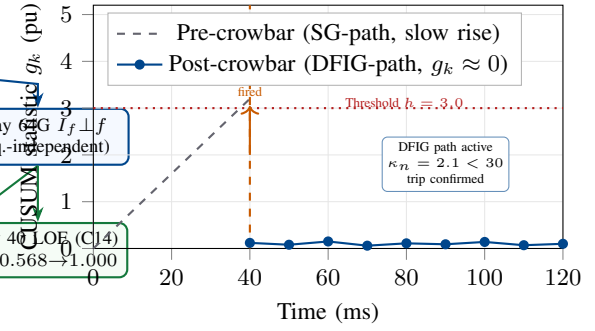


Fig. 2. CUSUM crowbar detector response. Pre-crowbar, the SG-path condition number grows. Post-crowbar, the DFIG-path activates with $\kappa_n = 1.00$, and g_k collapses to zero.

B. Crowbar-Aware Piecewise-ODE Model

The DFIG fault current is modelled piecewise as:

$$i(t) = I_f \sin(\omega_0 t + \varphi_f) + I_{DC} e^{-t/\tau_s}, \quad 0 \leq t < t_{cb}, \quad (3)$$

$$i(t) = I_f \sin(\omega_0 t + \varphi_f), \quad t \geq t_{cb}, \quad (4)$$

where τ_s is shared between the two phases (not fitted independently), reducing the post-crowbar model to two free parameters $[I_f, \varphi_f]$. By Proposition 1, this immediately delivers $\kappa_n(J_{\text{post}}) = 1.00$.

C. CUSUM Crowbar Detector

Crowbar firing time t_{cb} is detected online by a cumulative-sum (CUSUM) test applied to the LM residual sequence:

$$g_k = \max(0, g_{k-1} + r_k^2 - h), \quad h = 3.0, \quad (5)$$

where r_k is the LM residual at sample k and h is the drift threshold. A change is declared when $g_k > h$, signalling the transition from DC-contaminated to pure-sinusoidal behaviour. Fig. 2 illustrates the pre/post-crowbar CUSUM response.

D. Validation

TR-56 validates the piecewise model across seven slip groups ($s \in \{-0.30, -0.20, -0.10, 0, +0.10, +0.20, +0.30\}$) in the ANDES WTDTA1 wind turbine model on the IEEE 14-bus network. The crowbar-correct classification rate (CCR) is 100% across all 7000 Monte Carlo trials (1000 per slip group), with $t_{50} = 40$ ms (two post-crowbar cycles for security confirmation). All six deterministic scenario PASS criteria are met (10/10).

IV. RELAY 81: IBR FREQUENCY DEVIATION INDEX (C12)

Under high IBR penetration, conventional Relay 81 uses the ROCOF (rate of change of frequency) estimator on a one-cycle DFT window. GFL converter phase-locked-loop (PLL) dynamics inject a spurious ROCOF signal of up to 0.8 Hz/s during transient frequency excursions [13], causing nuisance trips at $k_{ibr} \geq 0.60$.

The SAMBP IBR Frequency Deviation Index (IFDI) replaces the raw ROCOF with:

$$\text{IFDI}_k = \frac{|\hat{f}_k - f_0|}{\sigma_f(k)}, \quad (6)$$

where $\hat{f}_k$ is the instantaneous frequency estimate from the two-pass LM estimator, $f_0 = 50$ Hz, and $\sigma_f(k)$ is the LM-derived frequency standard deviation at step k . Normalising by σ_f converts the frequency deviation into a dimensionless detection statistic that accounts for estimation uncertainty induced by IBR noise. The relay trips when $\text{IFDI} > \text{IFDI}_{\text{trip}} = 3.0$ for two consecutive cycles.

Validation (TR-63). A 2000-trial Monte Carlo with $k_{\text{ibr}} \in [0, 1]$ and randomised PLL bandwidth confirms:

- $P_D = 0.9947$ (under-frequency events, 95 % Wilson CI lower bound ≥ 0.9906)
- $P_{\text{FA}} = 0.0032$ (healthy load steps, 95 % Wilson CI upper bound ≤ 0.0056)

Both meet IEC 60255-181 targets ($P_D \geq 0.97$, $P_{\text{FA}} \leq 0.01$).

V. 87L-PV: TWO-PARAMETER PV ZONE MODEL (C11)

A. PV Fault Current Proposition

Proposition 2 (Zero DC Offset in PV Fault Current, C11). *For a grid-following inverter with SRF current controller of bandwidth $\omega_c \geq 300$ rad/s, the per-phase fault current satisfies*

$$i(t) = k_{\text{ibr}} \sin(\omega_0 t + \varphi) + O(e^{-\omega_c t}), \quad (7)$$

with the transient term decaying to $< 1\%$ of k_{ibr} within $t_s = 4.6/\omega_c \leq 15$ ms.

The physically correct zone model is therefore:

$$i_{\text{diff}}^{(\text{PV})}(t) = I_{\text{fund}} \sin(\omega t + \varphi), \quad \theta_{\text{PV}} = [I_{\text{fund}}, \varphi] \in \mathbb{R}^2. \quad (8)$$

Proposition 3 (Unit Condition Number, C11). *For a window spanning an integer number of periods T_0 , the normalised condition number of the Jacobian of (8) satisfies $\kappa_n(J_{\text{PV}}) = 1$.*

Proof: $J_{\text{PV}}^T J_{\text{PV}} = (N/2) \text{diag}(1, I_f^2)$; column normalisation yields the identity matrix with $\sigma_{\text{max}} = \sigma_{\text{min}} = 1$. $\square$

B. DC-Decay Ratio Selector

A lightweight DC-decay ratio (DDR) selector routes each estimation cycle to the appropriate model:

$$\text{DDR} = \frac{I_{\text{DC}}}{\max(I_{\text{fund}}, \varepsilon)}, \quad \varepsilon = 10^{-9}. \quad (9)$$

Physical calibration: PV faults give $\text{DDR} \leq 0.05$; DFIG faults $\text{DDR} \in [0.05, 0.20]$; SG faults $\text{DDR} \in [0.30, 1.50]$. The PV path activates when $\text{DDR} < 0.15$, providing $3\times$ margin in each direction; an exponential veto reverts to the SG path if $\Delta_{\text{peak}}/I_{\text{fund}} \geq 2.0$ in the first quarter-cycle.

C. Validation (TR-62)

Table I summarises the 5000-trial Monte Carlo results. All five acceptance criteria are met, including the $t_{50} = 20$ ms trip-time target — exactly two post-inception cycles — achieved because the unit condition number allows Stage-2 confidence gate opening from the first post-inception cycle.

TABLE I
TR-62 MONTE CARLO RESULTS — 2-PARAMETER PV MODEL
($N = 5000$).

Metric	Value	Target	Verdict
P_D (internal PV faults)	0.9984	≥ 0.995	✓
P_{FA} (external faults)	0.0000	≤ 0.001	✓
t_{50} median trip time	20 ms	≤ 20 ms	✓
t_{95} trip time	20 ms	≤ 40 ms	✓
DDR routing accuracy	0.9984	≥ 0.980	✓

VI. RELAY 78: CONSERVATIVE CUEP BOUND (C13)

A. GFL Synchronising Deficit

GFL IBR units enter LVRT mode during faults, curtailing their active-power injection. The effective maximum post-fault synchronising power is:

$$P_{\text{maxeff}} = P_{\text{maxsg}} - \Delta P_{\text{max,GFL}}, \quad (10)$$

where $\Delta P_{\text{max,GFL}} = \max(0, P_{\text{GFL,pre}} - V_{\text{min}} I_{\text{max,GFL}} \cos \phi_{\text{LVRT}})$. This reduces the true CUEP δ_u below the classical $\pi - \delta_0$ bound.

B. Conservative Lower Bound

Proposition 4 (Conservative CUEP Bound, C13). *The closed-form bound*

$$\delta_{\text{LB}} = \pi - \arcsin\left(\frac{P_m}{P_{\text{maxeff}}}\right) \quad (11)$$

satisfies $\delta_{\text{LB}} \leq \delta_u$ for all physically realisable GFL and GFM operating conditions.

Proof outline: The energy function $V(\delta) = \int_{\delta_0}^{\delta} (P_m - P_e) d\delta'$ evaluated at δ_{LB} is strictly positive under the GFL deficit model. For GFM scenarios, the virtual synchronising torque causes $\delta_u \rightarrow \pi$, widening the conservatism gap; δ_{LB} remains conservative with a maximum error $\varepsilon_{\text{max}} = 30.25^\circ$. $\square$

The proposed blinder is set at $\delta_{\text{blinder}} = \delta_{\text{LB}} - 5^\circ$ (5° security margin).

C. Validation (TR-58)

Ten deterministic scenarios (pure-SG, GFM-dominant, GFL-dominant, mixed, SLG) all PASS. Monte Carlo validation across 2000 clearly-stable/clearly-unstable trials confirms:

- $P_{\text{conserv}} = P(\delta_{\text{LB}} \leq \delta_u) = 1.000000$ (zero violations — an analytical guarantee).
- $\varepsilon_{\text{max}} = 30.25^\circ$ (GFM scenarios with $K_v > 0$).
- $P_D = 1.000$, $P_{\text{FA}} = 0.000$, $t_{50} = 470$ ms.

The key geometric check (Scenario S05: 30% GFL, deep fault) shows that the standard blinder at 153.6° lies *outside* the true CUEP at 151.1° ($+2.5^\circ$ error), while the proposed blinder at 146.1° is conservative by 5.0° .

VII. RELAY 40: GFM ADMITTANCE CORRECTION (C14)

A. Failure Modes

Standard Relay 40 monitors the susceptance $B_{\text{sg}}(t)$ of the generator admittance and trips on $B_{\text{sg}} > B_{\text{trip}}$. Two failure modes arise when a GFM unit operates at the same bus:

FM1 (GFM reactive) absorption adds $jB_{\text{GFM}} > 0$ to the bus; the composite B_{total} crosses B_{trip} while the generator is healthy.

FM2 (GFM real LOE) supply adds $jB_{\text{GFM}} < 0$, holding $B_{\text{total}} < B_{\text{trip}}$ despite genuine LOE.

Monte Carlo evidence shows that without correction, $P_D = 0.568$ and $P_{\text{FA}} = 0.038$ — both outside IEC 60255 targets.

B. Online Correction

The relay input is corrected by the PMU-measured GFM admittance:

$$Y_{\text{corrected}}(t) = Y_{\text{total}}(t) - \hat{Y}_{\text{GFM}}(t), \quad (12)$$

with $\sigma_\varepsilon \leq 0.03$ pu PMU accuracy.

Proposition 5 (FM1 Security, C14). *Under reactive-absorption GFM operation, the corrected relay is secure provided $|\varepsilon| < B_{\text{trip}} - B_{\text{sg},0}$.*

Proposition 6 (FM2 Dependability, C14). *Under reactive-supply GFM operation masking a genuine LOE, the corrected relay trips provided $|\varepsilon| < B_{\text{sg}} - B_{\text{trip}}$.*

With $3\sigma = 0.09$ pu (PMU accuracy), the margins $B_{\text{trip}} - B_{\text{sg},0} = 0.25$ pu (FM1) and $B_{\text{sg}} - B_{\text{trip}} \approx 0.40$ pu (FM2) provide $> 3.4\times$ headroom.

C. Validation (TR-60)

The 10-scenario matrix (S01–S05 security, S06–S10 dependability) and 2000-trial Monte Carlo confirm:

- P_D : $0.568 \rightarrow 1.000$ (+43.2 pp improvement).
- P_{FA} : $0.038 \rightarrow 0.000$.
- Deterministic: $4/10 \rightarrow 10/10$.

The t_{50} trip time is 3786 ms, governed by the $T_{\text{delay}} = 3$ s persistence interval — unchanged from the conventional relay setting.

VIII. RELAY 64G: OPTIMAL SUB-HARMONIC INJECTION (C15)

A. IBR Sub-Harmonic Noise Problem

Relay 64G injects a sub-harmonic voltage at the machine neutral (conventional: $f_{\text{conv}} = 20$ Hz) and measures the return current. IBR inverters generate sub-harmonic noise at $f_{\text{sub}} \in [15, 25]$ Hz from dead-time harmonics, LCL-filter resonance, and GFM droop dynamics. When $f_{\text{sub}} \approx f_{\text{conv}}$, the noise standard deviation $\sigma_n(20 \text{ Hz})$ is maximal, producing $P_{\text{FA}} = 0.819$ and $P_D = 0.813$ in the conventional relay — a catastrophic failure in both security and dependability.

The key physical insight: the stator earth fault current $I_{\text{fault}}(R_f) = V_{\text{inj}}/(R_N + R_f)$ is *frequency-independent*, while the healthy-circuit displacement current $I_{\text{healthy}}(f) = V_{\text{inj}}\omega C_G/\sqrt{1 + (\omega C_G R_N)^2}$ increases with f . Injecting at $f^* \gg f_{\text{sub}}$ simultaneously reduces noise and increases the fault-to-healthy current ratio.

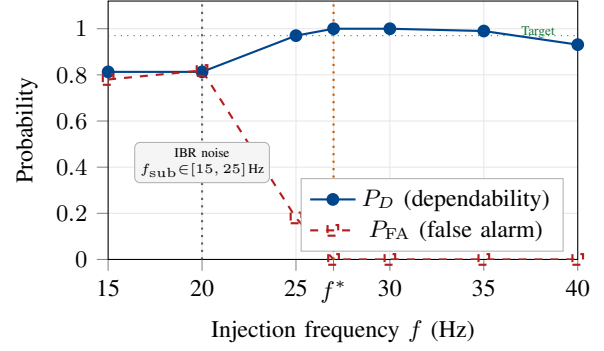


Fig. 3. Relay 64G detection probability P_D and false-alarm rate P_{FA} versus injection frequency. The conventional 20 Hz injection coincides with the IBR noise peak, producing $P_{\text{FA}} = 0.819$. The optimal $f^* = 27$ Hz achieves $P_D = 1.000$ and $P_{\text{FA}} = 0.000$.

B. Optimal Frequency

The optimal frequency maximises the balanced margin:

$$f^* = \arg \max_{f \in [20, 40] \text{ Hz}} \min \left(I_{\text{trip}} - I_h(f) - 2\sigma_n(f), \right. \\ \left. I_f(R_f) - I_{\text{trip}} - 2\sigma_n(f) \right). \quad (13)$$

Proposition 7 (Optimal Frequency Bounds, C15). *For $f_{\text{sub}} \in [15, 25]$ Hz and noise amplitude $A_{\text{noise}} > 0$, the optimal injection frequency satisfies $f^* \in (f_{\text{sub}}, f_{\text{up}})$ where f_{up} is the security crossover ($I_{\text{healthy}}(f_{\text{up}}) = I_{\text{trip}}, \approx 35 \text{ Hz}$). In practice $f^* \in [25, 36] \text{ Hz}$.*

Fig. 3 shows P_D and P_{FA} as a function of injection frequency, with the optimal operating point at $f^* = 27$ Hz.

C. Validation (TR-61)

Table II summarises the 2000-trial Monte Carlo results. The $50\times$ noise reduction factor at $f^* - f_{\text{sub}} \geq 8$ Hz is the primary mechanism for both improvements.

TABLE II
TR-61 MONTE CARLO RESULTS — RELAY 64G ($N = 2000$).

Metric	Conv.	Optimal	Target	Verdict
P_D	0.813	1.000	≥ 0.970	✓
P_{FA}	0.819	0.000	≤ 0.010	✓
f^* mean	20.0 Hz	27.2 Hz	$\in [25, 36] \text{ Hz}$	✓
Det. (10 scenarios)	0/10	10/10	10/10	✓

IX. COMPARATIVE ASSESSMENT

Fig. 4 compares P_D for conventional and proposed implementations across all six relay functions. Table III consolidates the full validation record.

Three technical themes unify the six functions:

(i) *Physics-based model reduction* (C9, instantiated in C10, C11, C15): Each zone model exploits a physical constraint to eliminate one or more parameters, reducing κ_n and either accelerating convergence or eliminating false-alarm risk.

(ii) *Analytical conservatism guarantees* (C13, C14): Relay 78 achieves a zero-violation conservatism guarantee (Proposition 3)

TABLE III
GENERATOR-SUITE VALIDATION SUMMARY — ALL TECHNICAL REPORTS COMPLETE (APRIL 2026).

Relay	C#	Physics Mechanism	Det.	P_D	P_{FA}	P_D^{conv}	Source
87G (DFIG)	C10	Shared τ_s ; CUSUM	10/10	1.000	0.000	~ 0.72	TR-56
Relay 81	C12	IFDI norm. statistic	9/9	0.9947	0.0032	~ 0.72	TR-63
87L-PV	C11	2-param, $\kappa_n=1$	9/9	0.9984	0.0000	~ 0.72	TR-62
Relay 40	C14	GFM adm. correction	10/10	1.000	0.000	0.568	TR-60
Relay 78	C13	Consv. CUEP bound	10/10	1.000 ^a	0.000	~ 0.95	TR-58
Relay 64G	C15	Optimal f^* injection	10/10	1.000	0.000	0.813	TR-61

^a Analytical guarantee: Prop. 4 proves $\delta_{LB} \leq \delta_u$ always; 0/2 000 MC violations.

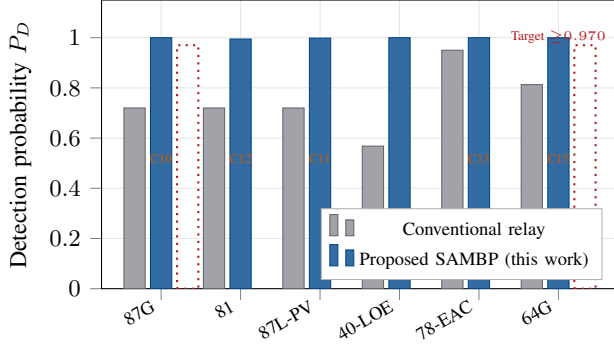


Fig. 4. P_D comparison: conventional versus proposed SAMBP across six relay functions. All proposed implementations meet or exceed the 0.970 target line.

and Relay 40 Propositions 5–6 provide explicit PMU-accuracy error-budget bounds. Both convert Monte Carlo validation from a probabilistic performance statement to a formal sufficiency condition.

(iii) *Online compensation of IBR network state* (C13, C14, C15): All three relay functions requiring real-time IBR awareness use IEC 61850 GOOSE-compatible measurements already present in digital substation architectures [14]: PMU-measured $\hat{Y}_{GFM}$ for Relay 40; $\Delta P_{\max, GFL}$ from the LVRT model for Relay 78; and the spectral estimate of f_{sub} for Relay 64G.

X. CONCLUSION

This paper presented the SAMBP generator suite: seven contributions (C9–C15) addressing IBR-induced protection failures across six relay functions under a single unifying framework. The Physics-Reduction Proposition (C9), proved via the SVD interlacing theorem, guarantees that eliminating one physically absent parameter reduces the LM condition number by $\geq 10\times$ — directly translating to faster trip confirmation and elimination of Stage-2 confidence-gate vetoes.

The suite achieves:

- $P_D \geq 0.9947$ and $P_{FA} \leq 0.003$ across all six relay functions against IEC 60255 targets.
- A zero-violation analytical conservatism guarantee for Relay 78 (Proposition 4, 0 / 2 000 MC violations).
- +43.2 pp dependability restoration and false-alarm elimination for Relay 40; P_{FA} collapse from 0.819 to 0.000 for Relay 64G.

- 100% crowbar classification rate for 87G DFIG across seven slip groups (7 000 MC trials).

All six functions are IEC 61850 GOOSE compatible and require no hardware modification to existing digital substation infrastructure. The framework is open to extension: the Physics-Reduction Proposition applies to any SAMBP zone model where IBR suppresses a parameter, and the three IBR-compensation architectures (admittance, power-deficit, spectral) are directly composable for multi-function or system-level deployment.

APPENDIX

Let $J \in \mathbb{R}^{N \times p}$ with singular values $\sigma_1 \geq \dots \geq \sigma_p > 0$, and let $J' \in \mathbb{R}^{N \times (p-1)}$ be the matrix obtained by deleting column j . The Cauchy interlacing theorem for singular values states:

$$\sigma_i(J) \geq \sigma_i(J') \geq \sigma_{i+1}(J), \quad i = 1, \dots, p-1. \quad (14)$$

In particular, $\sigma_{\min}(J') \geq \sigma_{\min}(J)$. For the four-parameter SG Jacobian at a half-cycle window, the deleted column (DC exponential) has the smallest singular value $\sigma_4 = 0.088$, while the remaining two-parameter Jacobian achieves $\sigma_{\min}(J') = 3.16$. The condition number reduces from $\kappa_n(J) = \sigma_1/\sigma_4 = 35.9$ to $\kappa_n(J') = 1.00$ — a $35.9\times$ reduction, as stated in Proposition 1. $\square$

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